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ORIGINAL ARTICLE

A Novel Proxy of Latent Rental Housing Demand: Evidence from US Markets

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Abstract

We introduce the Rental Density Index (RDI)—a metric that captures the household density of rental units—as a scalable and behaviorally grounded proxy for latent rental housing demand. Traditional demand indicators like occupancy or absorption are bounded by supply and often fail to reflect underlying crowding pressure. In contrast, changes in RDI (ΔRDI) reveal when renters are consolidating space, signaling excess demand, or spreading out, signaling slack. Using panel data for the 200 largest U.S. metropolitan areas from 2005 to 2023, we show that ΔRDI robustly predicts future rent growth. A two-stage least squares model, instrumenting for RDI growth with plausibly exogenous foreign in-migration shocks, reveals a positive and statistically significant causal effect on next-year relative rent growth. Out-of-sample tests further show that RDI-based forecasts outperform ARIMA and naïve models across one-, five-, and ten-year horizons. Event studies and regime classifications confirm that crowding transitions are consistently followed by directional rent changes. The Rental Density Index provides a simple yet powerful tool for identifying housing market tightness, classifying supply-demand regimes, and forecasting rent performance. It is especially useful in contexts where traditional demand indicators are constrained or unavailable.

KEYWORDS
density, supply and demand, multifamily rent-growth, apartment markets

1 | INTRODUCTION

Housing shortages and affordability concerns have risen to the forefront of policy debates in major urban markets. In the United States, housing production has consistently lagged population growth for decades, contributing to an estimated national shortfall of 4.4 million housing units (Betancourt, 2022). Nearly half of U.S. renter households now spend over 30% of their income on housing (Bureau, 2022), and from 2000 to 2024, the consumer price index (CPI) for shelter exceeded the CPI for all other items by 30% (of St. Louis, 2024). At the same time, select high-growth regions have recently experienced rent declines due to a glut of new supply (Mott, 2024). This juxtaposition of chronic national undersupply with localized oversupply underscores a deeper issue: the lack of a reliable metric to measure consumer housing demand.

Traditional indicators of demand in multifamily real estate, such as occupancy rates and net absorption, are informative but fundamentally supply-constrained. Occupancy rates are naturally bounded at 100%, and absorption cannot exceed the rate of new deliveries (Mueller & Laposa, 1999). These constraints obscure excess demand: when all available and affordable units are occupied, latent demand becomes invisible to market participants and researchers alike (Gabriel & Nothaft, 2001; Sirmans, Sirmans, & Benjamin, 1991; Pyhrr, Cooper, & Wofford, 1999). This problem inhibits clear attribution of rent increases to supply versus demand dynamics (Pennington, 2021; Molloy, 2022). Moreover, ongoing academic debate persists around whether rising rents in constrained markets result more from supply-side limitations (Saiz, 2010) or from heightened demand for desirable locations (Davidoff, 2015). Without a transparent, consistent demand-side metric, attempts to assess equilibrium conditions remain incomplete.

Abbreviations: MSA, Metropolitan Statistical Area; RDI, relative density index;

In this paper, we introduce a novel empirical measure of rental housing demand: the *rental density index* (RDI), defined as the average number of occupants per rental unit in a given metropolitan area. Unlike occupancy or absorption, RDI is not inherently bounded and therefore can reflect intensifying demand even in fully occupied markets. The conceptual foundation is straightforward: as rents rise, renters economize on space—whether by delaying household formation, taking on roommates, or crowding—thus increasing the ratio of people per rental unit and signaling pent-up demand. By observing shifts in RDI over time, we capture underlying demand pressures that traditional metrics obscure.

Present trends support this reframing of demand, as density, rental prices, and rental preferences have all shifted meaningfully. For most of its history, the U.S. was less densely populated than its high-income peers, but in 1992 this changed. In that year, the US grew to 28 people per square kilometer, outpacing the High Income Nations whose average was 23 people per square kilometer (Bank, 2025). The most recent figures report the US with a density of 36 and the High Income Nations with an average density of 27. Over this same period, rental prices have outpaced both wages and non-shelter inflation (Feiveson et al., 2024; of St. Louis, 2024), with younger cohorts increasingly preferring rental housing (Mae, 2023).

Prior literature in urban economics and real estate has studied density extensively, often linking increased geographic density to higher wages, rents, and productivity due to agglomeration effects (Titman, Wang, & Yildirmaz, 2024; Liu, Pan, & Li, 2018). However, these studies generally define density in terms of spatial density: people per square mile or apartments per square mile. Our focus differs: we define density as the quotient of population over rental units, which provides a clearer window into the number of people actively competing for housing. Unlike geographic density, this measure is directly responsive to shifts in demand, especially during periods of supply constraint or shocks.

Our framework builds on the standard assumption that, all else equal, households derive higher utility from larger living spaces (Muth, 1969; Molloy, 2022). In high-rent markets, however, the extra price per square foot may eventually exceed the marginal benefit of additional space. To economize, renters may respond by sharing units (adding roommates), downsizing to smaller apartments, or relocating to less expensive regions. Conversely, when real rents fall or new supply comes online, the marginal cost of extra space might drop below its marginal utility, enticing households to spread out, move to larger units, or reduce unitshare.The aggregate effect of these choices combined with the new supply additions or subtractions determines the direction of the Rental Density Index change. Because the RDI directly captures these space–consumption margins, year–over–year changes in RDI are proxies for shifts in aggregate housing demand at prevailing prices.

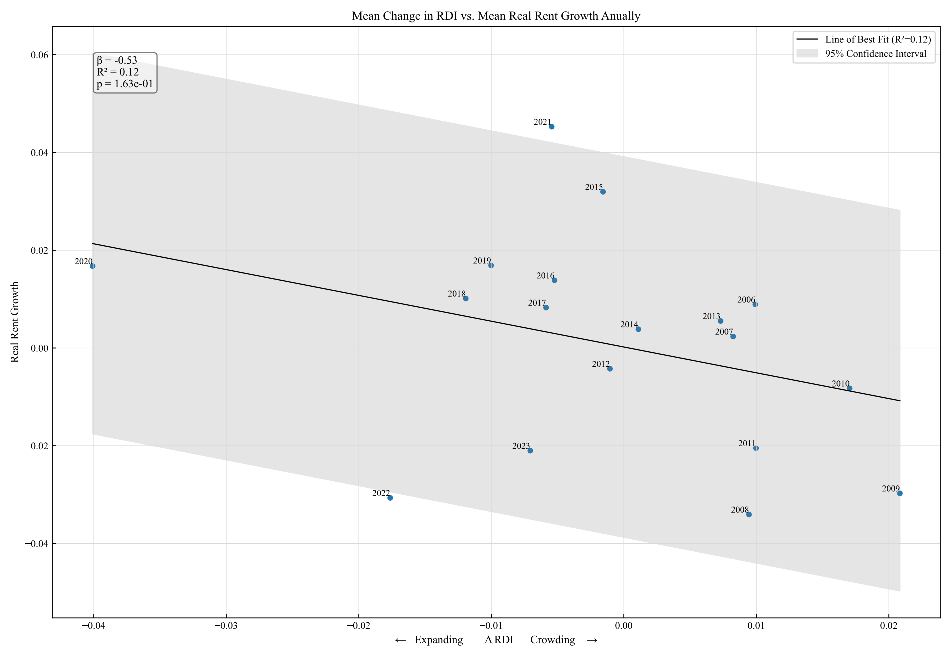


FIGURE 1 Relationship between average annual change in Rental Density Index (horizontal) and average annual relative real rent growth (vertical) in the 200 largest MSAs for each year between 2005 and 2023.

Focusing on a panel of the 200 largest metropolitan statistical areas (MSA) between 2005 and 2023, we calculate the RDI. It ranges from 1.79 occupants—per–rental to 3.87. The value itself is telling. MSAs with RDI greater than the median experience significantly higher next—year real rent growth than those with RDI values below the median.

And while the RDI is useful as a spot metric, so too is the change in RDI. A positive change in RDI (densification) means that the number of occupants—per—rental is increasing; conversely a negative change in RDI (de-densification) corresponds to a decrease in occupants—per—rental. The reasons for an increasing RDI are, in most cases, population growth in excess of supply growth. However, it could be possible that housing units declined as population remained static. Although the RDI does not explicitly factor in the count of households, because household count grows almost monotonically, the variable is still robust.

This empirical strategy corresponds with consistent rent growth differences across multiple time horizons. Over the one-year horizon, positive RDI growth markets significantly underperform negative RDI growth markets. That is, the densifying markets underperform the de-densifying markets: as renters consolidate households, rent declines. The RDI value itself /(not the year-over-year change) has the opposite dynamic: markets with higher RDI values significantly outperform markets with lower RDI values. While this appears contradictory, it is consistent: a city that densifies has is willing to pay for fewer housing units per person that year, but demand is building, as cities cannot densify forever. And a city that has greater-than-average renters—per–unit has greater demand, and tends in the long-run toward a lower density, consuming more housing units as it unwinds. Akron is an illustrative example: from 2005 to 2011 it increased its occupants-per-rental from 2.06 to 2.32; during this time its real rent declined by 12.4%. In 2012 it began unwinding its pent-up demand, and in the years between 2013 and 2021 its rent increased by 9.9% as its RDI declined to 2.01.

Combining these two metrics: RDI and its change provides further insights. Taken together, they reveal when supply will be accretive to rent versus dillutive. When markets experience supply spikes, those with high RDI and negative RDI growth see rent grow in the following years. Conversely, markets with low RDI and positive RDI growth experience negative rent growth in times of high supply. These findings suggest that density-based classifications have predictive power and reflect latent demand more accurately than traditional indicators alone.

Our paper makes three principal contributions. First, we introduce the Rental Density Index (RDI), a novel, supply-unbounded metric of housing demand that can be computed at scale from readily available population and unitstock data. Second, we show the RDI is a significant indicator of latent demand, using a two-stage least squares model and a very broad panel of foreign in-migration data to measure demand shocks. Finally, we demonstrate RDIs empirical value using a suite of use-cases applicable to investors, renters, developers, and policy-makers.

These findings carry immediate applications for all housing—market stakeholders. For policymakers, RDI functions as an early-warning gauge of emerging shortages, enabling targeted zoning reforms, calibrated subsidy programs, or expedited permitting in precisely those submarkets under the greatest pressure. Institutional investors and residential developers can embed RDI trends into feasibility and risk models, avoiding the twin hazards of overbuilding in cooling metros or underbuilding in tightening ones. Finally, RDI sharpens the housingaffordability debate by identifying the locales where rent increases are most likely to outstrip income growth, thereby guiding tenant–protection policies, rent–assistance programs, and other affordability interventions to the communities that need them most.

In the next section we discuss research on other measures of demand before presenting details on our proposed variable. The Data and Descriptive Statistics section describes and analyzes the density data we used, while the Empirical Strategy section presents evidence and illustrates statistical tests performed to evaluate the validity of the econometric model. In the Forecasting and Validation section, we evaluate the RDI in five different scenarios, showing its power in rent-growth forecasting. The final sections discuss the results of our tests before concluding with implications and further research suggestions.

2 | BACKGROUND AND LITERATURE REVIEW

A substantial literature in urban and housing economics has sought to measure rental demand using structural modeling, hedonic estimation, and affordability thresholds. Price elasticity estimates, often based on income and housing cost shares, provide insight into how demand responds to price movements under equilibrium assumptions (Green, Malpezzi, & Mayo, 2002a; Malpezzi & Green, 1996b). Hedonic models explain rent levels as a function of unit characteristics and neighborhood amenities, while housing affordability metrics attempt to measure stress via cost-to-income ratios. Though valuable, these approaches often require microdata that are unavailable across time and space or assume frictionless adjustment, limiting their utility in settings with short-run disequilibrium or constrained inventory.

The multifamily housing sector has traditionally relied on indicators such as occupancy rates and net absorption to characterize demand. These measures are intuitive and operationally convenient but are constrained by the size of the existing rental stock. Occupancy, by definition, cannot exceed 100 percent, and absorption is mechanically bounded by the number of new units delivered (Mueller & Laposa, 1999; Gabriel & Nothaft, 2001). As a result, such metrics are poorly equipped to detect latent demand, specifically periods in which prospective renters are unable to form households or must cohabitate due to limited supply or rising costs (Sirmans et al., 1991; Pyhrr et al., 1999). This limitation has become more acute in recent years as algorithmic revenue management systems increasingly target stabilized occupancy thresholds (for example, 95 to 96 percent), further muting variation and diminishing their informational content (CalderWang & Kim, 2024).

While these indicators retain short-run forecasting value, they offer limited theoretical insight into renter behavior under constraint. Several studies have documented rent growth during periods of high occupancy but tend to attribute this to restricted supply rather than suppressed household formation (Goodman, 1992; Wheaton, 1991). Structural models, including utility-based choice models (Rosenthal, 1997a) or equilibrium migration frameworks, provide more detailed behavioral inference but require data on preferences, incomes, or location-specific amenities that are rarely standardized across markets.

Parallel literature in urban economics has studied density, typically defined as people or dwellings per square mile, as a proxy for agglomeration or spatial efficiency. Seminal work by (Glaeser et al., 2001) and (Duranton & Puga, 2004) links higher density to economic productivity, while more recent studies explore the dual role of densification in shaping housing affordability and urban form (Ahlfeldt & Pietrostefani, 2019; Albouy et al., 2015). Yet geographic density is primarily a land-use metric; it does not directly reflect household formation or space consumption behavior.

Our work reframes density through a behavioral lens by defining it as the number of people per occupied rental unit: the Rental Density Index (RDI). This metric captures demand pressure more directly than geographic density by tracking how renters adjust their living arrangements under price or supply stress. If renters prefer more space but are observed to cohabitate as rents rise or supply tightens, an increase in RDI reflects a behavioral response to constrained conditions. Conversely, declines in RDI suggest slack: households can afford to spread out. Unlike traditional metrics, RDI growth (ΔRDI) enables analysts to detect crowding pressures even in fully occupied markets.

RDIs strength lies in its scalability and interpretability. Constructed from widely available data — population, and rental inventory — it offers a transparent, repeatable measure of latent demand across markets and over time. This approach is especially valuable in contexts where elasticity models fail to identify pressure due to equilibrium assumptions or unavailable microdata. While elasticity studies estimate how much demand shifts in response to price, RDI captures whether pressure exists at current prices through observed space consumption.

In sum, our contribution is to reinterpret a familiar spatial concept as a dynamic, renter-centered metric that reflects real-time demand conditions. By linking changes in RDI to rent performance, we introduce a new tool for housing market analysis that complements traditional models and offers practical value for forecasting, segmentation, and policy design.

3 | DATA AND DESCRIPTIVE STATISTICS

The data used to calculate RDI is from the Public Use Microdata Statistics (PUMS). The PUMS data is an extremely valuable offering of the U.S. Census in conjunction with the American Community Survey. The data allow for visibility into myriad aspects of a respondent's life (including the number of people in their household and whether they rent or own their home), annonymized for privacy and standardized to create a comprehensive view of the entire country. Our market rent data and inventory growth data is sourced from Costar. This includes nominal rent per square foot and inventory units. Our inflation measures come from the U.S. Bureau of Labor Statistics. And our foreign immigration data at the market level comes from the US Census. We restrict our analysis to the 200 largest U.S. metropolitan statistical areas (MSAs) by multifamily inventory as of 2005, ensuring robust sample representation and consistency over time.

To normalize pricing data across time, we convert all nominal rent figures into real terms using the Consumer Price Index for All Urban Consumers (CPI-U), all-items, provided by the BLS. We convert nominal rent-growth figures into real rent-growth figures by subtracting the annual CPI-U change from the annual rent per square foot change. Although MSA-level inflation indices excluding rent would be ideal, such data are unavailable at a consistent and granular level for our study period. Therefore, real effective rent per square foot and rent growth metrics are adjusted using national CPI.

Importantly, we standardize rent-growth subtracting each year's median rent growth. We do this to remove the impact of national macroeconomic shocks (i.e., COVID-19). This also serves to stationarize the data and remove the autoregressive tendencies of timeseries data.

Our novel demand variable, the Rental Density Index (RDI), is calculated by summing the number of people living in rental units within an MSA, and dividing that by the sum of rental units within an MSA. It does not include vacant dwellings or seasonal dwellings. We compute its year-over-year percentage change—ΔRDI—to reflect demand dynamics more accurately. This measure avoids the bounded structure of traditional demand indicators like occupancy and absorption, which are upper bounded by 100%.

We exclude New Orleans, LA from our analysis because of its dynamics around the unfortunate aftermath of Hurricane Katrina. Within two years, this MSA had the greatest and least values of RDI growth as well as the greatest and least values of real rent change.

TABLE 1 Key variables used in the empirical analysis.

Variable	Description
`CPI`	Consumer Price Index for All Urban Consumers: All Items in U.S. City Average
`real_rentpsf`	Costar's Rent Per Square Foot divided by indexed `CPI`
`real_rent_growth`	Costar's Effective Rent Growth 12M less Annual percent change in `CPI`
`relative_real_rent_growth`	`real_rent_growth` less that year's median `real_rent_growth`
`inventory`	Costar's total number of rental units in an MSA
`rental_households`	Dwellings that are being rented within an MSA
`rental_population`	The number of people living in rental dwellings within an MSA
`pop`	Total MSA population
`RDI`	`rental_population` divided by `rental_households`
`delta_RDI`	Yearoveryear percentage change in `RDI`
`supply_growth`	Yearoveryear percentage change in `inventory`
`foreign_migration`	Foreign net migration divided by the area's population

Rental Density Index Construction

The PUMS data is large, and complex, though very well documented. The output of the raw data requires parsing to aggregate to the MSA level. This section describes that aggregation process.

1. Grouping Records

Records are grouped by survey year (`YEAR`), household identifier (`SERIAL`), metropolitan area name (`MET_NAME`), group quarters status (`GQ`), and ownership status (`OWNERSHP`). For each group i, we compute:

$$\text{POPULATION}_i = \sum_{j \in i} \text{PERWT}_j \tag{1}$$

$$\text{HOUSEHOLDS}_i = \text{HHWT}_{j^*} \tag{2}$$

where PERWT_j is the person weight for individual j, and HHWT_{j^*} is the household weight taken from the first individual in the group.

2. Ownership Label Mapping

Ownership status is translated from numeric codes to descriptive labels as follows:

$$\text{OWNERSHP} = \begin{cases} \texttt{"OWNED"} & \text{if code} = 1 \\ \texttt{"RENTED"} & \text{if code} = 2 \\ \texttt{"OTHER"} & \text{otherwise} \end{cases} \tag{3}$$

3. Aggregation by Metro and Year

We then re-aggregate the grouped data by year y, metro area m, and ownership category o:

$$\mathrm{POPULATION}_{y,m,o} = \sum_{i \in (y,m,o)} \mathrm{POPULATION}_i \tag{4}$$

$$\mathrm{HOUSEHOLDS}_{y,m,o} = \sum_{i \in (y,m,o)} \mathrm{HOUSEHOLDS}_i \tag{5}$$

4. Pivoting and Computing Totals

We pivot the data to obtain separate columns for each ownership category and compute totals:

$$\mathrm{TOTAL\_POPULATION}_{y,m} = \mathrm{POP\_OWNED}_{y,m} + \mathrm{POP\_RENTED}_{y,m} + \mathrm{POP\_OTHER}_{y,m} \tag{6}$$

$$\mathrm{TOTAL\_HOUSEHOLDS}_{y,m} = \mathrm{HH\_OWNED}_{y,m} + \mathrm{HH\_RENTED}_{y,m} + \mathrm{HH\_OTHER}_{y,m} \tag{7}$$

5. Density Calculation

Finally, population-to-household densities are calculated per ownership type:

$$\mathrm{DENSITY\_OWNED}_{y,m} = \frac{\mathrm{POP\_OWNED}_{y,m}}{\mathrm{HH\_OWNED}_{y,m}} \tag{8}$$

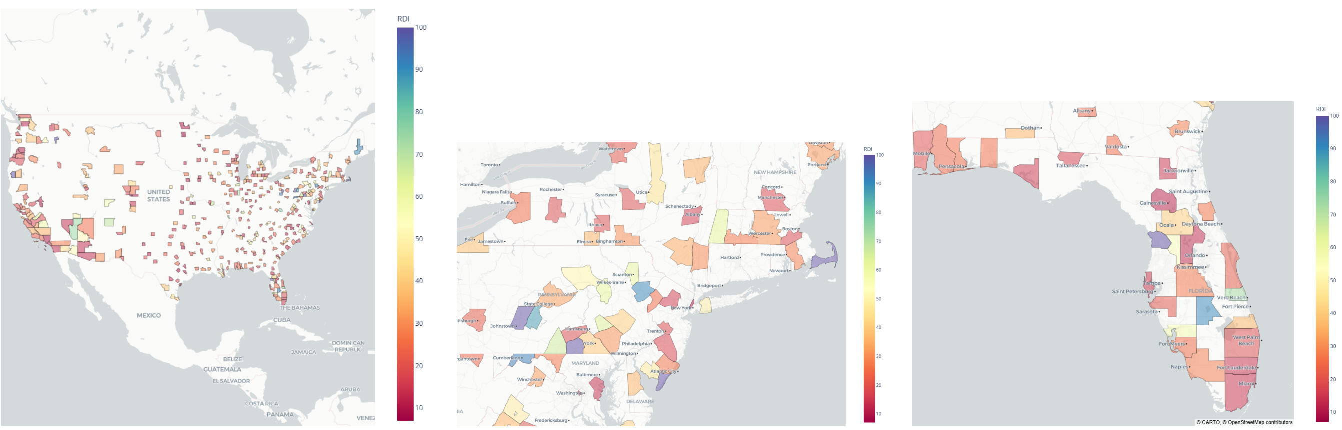
$$\mathrm{DENSITY\_RENTED}_{y,m} = \frac{\mathrm{POP\_RENTED}_{y,m}}{\mathrm{HH\_RENTED}_{y,m}} \tag{9}$$

$$\mathrm{DENSITY\_OTHER}_{y,m} = \frac{\mathrm{POP\_OTHER}_{y,m}}{\mathrm{HH\_OTHER}_{y,m}} \tag{10}$$

The variable DENSITY_RENTED becomes our so-called Rental Density Index.

Although we do not use the OWNED or OTHER variables, we compute them for thoroughness and for future use and make them available with our code.

Within the 200 MSAs in the study,the RDI ranges from 1.68 (Bloomington, IN; 2023) to 3.74 (Provo, UT; 2009).



(a) United States

(b) Tri-State area

(c) Florida

FIGURE 2 Rental-Density-Index (RDI) choropleths at three spatial scales.

The RDI itself is useful for identifying how tight a market is at a given point in time, and its absolute level is shaped by long-run demographic and structural trends in rentership and household formation. It is rare for a market to shift more than one half of one person-per-rental-dwelling, even over 20 years. To aid in short-term insight and allow comparability across markets, we use the *year-over-year change in RDI*:

$$\Delta \mathrm{RDI}_{it} = \mathrm{RDI}_{it} - \mathrm{RDI}_{it-1}.$$

The change in RDI (ΔRDI) offers several advantages. First, it mitigates issues of nonstationarity in the level RDI, enabling cross-market comparison. Second, it reduces the impact of varying rentership percentages across cities. Most importantly, ΔRDI captures demand pressure on housing stock: when it rises, it indicates more people are consolidating into fewer rental units, signaling a preference for cohabitation over paying prevailing rents. When it falls, it implies renters are spreading out and absorbing more space, suggesting household formation and rents below-or-at utility value.

It is useful to distinguish between two sources of crowding pressure captured by ΔRDI: price-induced and supply-induced. Price-induced crowding occurs when rising rents outpace incomes, forcing renters to share space or delay household formation—behavioral responses that directly elevate RDI. Supply-induced crowding, by contrast, may arise even without rent escalation if housing stock fails to expand alongside population, resulting in more residents per unit out of sheer constraint. While these mechanisms are observationally similar, our framework implicitly captures both, interpreting ΔRDI as the revealed outcome of frictions between housing availability and household spatial preferences. In this way, a high (but not growing) RDI reflects pent-up demand pressures regardless of whether they arise from affordability shocks or constrained supply.

3.1 | Summary Statistics

Table 2 provides descriptive statistics for the key variables across the full panel. On average, the RDI across all MSAs and years is 2.4 persons per rental unit, with a standard deviation of 0.3. The average real rent per square foot is $0.85, and the average annual growth in rental supply is 1.78%. The missing records exist only in the beginning or ending years as the prior period and next-period values do not exist. Note the mean value of real_rent_growth is zero. Nationally, the Consumer Price Index for rent-of-shelter has exceeded the Consumer Price Index excluding shetler, thus the mean national real rent growth is not zero. The reason the mean is zero in these summary statistics owes to weighting. The CPI is weighted to be representitive of the US population. Our summary statistics are not weighted, and thus a high-population metro has the same weight (in calculating this mean) as the smallest metro. This is not a concern, as the data is accurate, and our analyses are focused on the MSA-level metrics.

TABLE 2 Summary Statistics

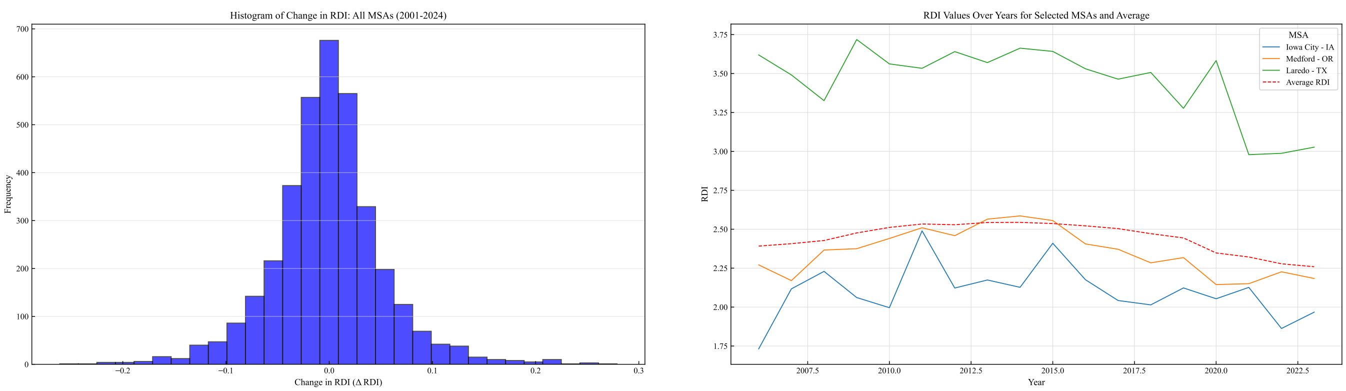
Variable	Count	Mean	Std	Min	25%	50%	75%	Max	Missing
`pop`	3800	1,113,385.26	1,778,446.92	106,878.20	254,907.63	487,778.16	1,067,312	14,897,850	0
`inventory`	3800	68,983.91	149,873.18	3,038	8,363	20,479.50	54,043.75	1,550,003	0
`supply_growth`	3600	0.0178	0.0252	-0.0978	0.00044	0.0104	0.0251	0.3211	200
`RDI`	3800	2.4432	0.3337	1.6857	2.2175	2.3810	2.5832	3.7417	0
`RDI_growth`	3600	-0.0012	0.0539	-0.2611	-0.0288	-0.0012	0.0251	0.2789	200
`real_rentpsf`	3800	0.8532	0.3102	0.4061	0.6589	0.7625	0.9485	2.7119	0
`real_rent_growth`	3800	0.0000	0.0295	-0.2744	-0.0180	-0.0012	0.0156	0.2738	0

3.2 | Coverage and Representativeness

The sample includes 3,600 MSA-year observations, covering a mix of large coastal cities, fast-growing Sun Belt metros, and slower-growing Midwestern regions. Together, these MSAs account for over 35 million rental dwellings, housing over 81 million residents, providing a true representative snapshot of national rental dynamics.

3.3 | Preliminary Observations

Figure 3 traces the national Rental Density Index (RDI) from 2006 to 2023. The average RDI over these 200 samples begins at 2.4 people per rental, rises to 2.5 in 2012, then declines toward 2.25 in 2023. A companion histogram of RDI is stationary but skewed slightly right, confirming the slow decline in RDI.



(a) A histogram of the annual change in RDI across 100 MSAs in the years 2001-2023. The outlier to the left is New Orleans in 2004, due to Huricane Katrina.

(b) A line graph showing the RDI values (population divided by apartment units) for the MSAs that were the min, max, and median in year 2001

FIGURE 3 Histogram of Change in RDI, and Line Graph of RDI

This pattern illustrates the need for a different demand metric, otherwise the concern for housing shortage seems to be contradicted by a monotonically decreasing housing density. Said differently, occupancy simply reports that units are full– it cannot distinguish whether they are full by choice (everyone prefers roommates) or by necessity (everyone must share). By contrast, year-over-year changes in RDI isolate the relative pace of new people versus new units. In the next section we investigate that property to reveal RDI as a proxy for latent demand.

4 | EMPIRICAL STRATEGY AND RESULTS

We estimate the causal effect of crowding, measured by the Rental Density Index (RDI), on future rent growth using a two-stage least squares (2SLS) instrumental variables (IV) approach. This strategy addresses the potential endogeneity of RDI, which may arise from reverse causality or omitted variable bias. For instance, higher rents may lead households to double up, mechanically increasing RDI, or unobserved local demand shocks could simultaneously drive both rent growth and household density.

4.1 | Instrumental Variable Justification: Share of Population Born in the MSA 20 Years Ago

To address potential endogeneity in the Rental Density Index (RDI), we instrument for RDI using the share of the MSA population that was born in the same MSA 20 years earlier. This variable reflects historical birth cohorts that are now reaching the typical age of independent household formation. Crucially, it is a historically anchored and plausibly exogenous proxy for future rental housing demand pressure.

Individuals typically begin to form independent rental households in their early 20s. Therefore, the size of the birth cohort from 20 years ago directly influences the intensity of household formation pressure today. A larger cohort implies that more individuals are approaching rental household formation age at time t, potentially increasing rental crowding if housing supply has not grown proportionately. As such, this instrument captures anticipated demographic-driven demand shocks that affect how many people are likely to seek housing relative to available rental stock.

While other approaches might use the contemporaneous share of 20-year-olds, that measure is potentially endogenous to current local conditions due to migration and reverse causality. In contrast, our measure is based on historical demographic eventsspecifically, the size of the cohort born in the MSA 20 years priorwhich are predetermined and unaffected by current-period shocks to rent or housing demand. This strengthens the instruments predictive validity for RDI while avoiding simultaneity bias.

By construction, the share of people born in an MSA 20 years ago is not influenced by current rental market conditions, as it depends on birth events and migration patterns two decades prior. These historical demographic dynamics are unlikely to be correlated with unobserved contemporary shocks to rent growth, satisfying the exclusion restriction. Importantly, we are not using the current presence of 20-year-olds (which may reflect selective migration), but rather the proportion of individuals currently living in the MSA who were born there two decades earlierthus minimizing concerns about endogenous responses to recent housing affordability or labor market conditions.

We bolster the validity of this instrument with two additional tests. First, a placebo test using contemporaneous rent growth as the outcome variable finds no significant effect, suggesting that the instrument influences rents primarily through its effect on RDI rather than through unobserved confounders. Second, a lagged instrument (based on the cohort born 21 years ago) produces consistent results, further reinforcing the temporal separation and the robustness of the causal identification.

4.2 | Two Stage Specification

The model is estimated on a panel of MSA-year observations, and includes MSA and year fixed effects to absorb time-invariant heterogeneity and national trends, respectively. The estimation equations are:

$$\text{Stage 1:} \quad \text{RDI}_{i,t} = \gamma_0 + \gamma_1 \cdot \text{Pct20yo}_{i,t} + \gamma_2 \cdot \mathbf{X}_{i,t} + \mu_i + \delta_t + u_{i,t}$$

$$\text{Stage 2:} \quad \text{RentGrowth}_{i,t+1} = \alpha_0 + \beta \cdot \widehat{\text{RDI}}_{i,t} + \alpha_1 \cdot \mathbf{X}_{i,t} + \mu_i + \delta_t + \varepsilon_{i,t}$$

Where:

- $\text{RentGrowth}_{i,t+1}$ is next-year real relative rent growth in MSA i,
- $\text{RDI}_{i,t}$ is average household size among renters,
- $\text{Pct20yo}_{i,t}$ is the instrument: percent of population age 20,
- $\mathbf{X}_{i,t}$ includes population growth and housing starts as controls,
- μ_i and δ_t are MSA and year fixed effects.

4.3 | Main Results

Table 3 presents the main 2SLS estimates using Pct20yo as the instrument. The coefficient on RDI is negative and statistically significant at the 1% level, consistent with the interpretation that increased crowdingvia larger households per rental unitexerts upward pressure on rents through constrained housing demand.

TABLE 3 Main IV Estimates: RDI and Next-Year Rent Growth

	Coef.	Std. Err.	p-value	95% CI
RDI	-0.0246	0.0091	0.0066	[-0.0424, -0.0069]
Population Growth	0.0096	0.0017	0.0000	[0.0063, 0.0130]
Housing Starts (%)	-0.2558	0.0608	0.0000	[-0.3750, -0.1366]
Constant	0.0608	0.0212	0.0040	[0.0194, 0.1023]
Observations	792			
F-statistic (first stage)	36.21			

4.4 | Placebo and Lagged Instrument Robustness

To test the exclusion restriction, we perform a placebo test using contemporaneous rent growth as the outcome. As shown in Table 4, the estimated effect of instrumented RDI on *same-year* rent growth is negative but statistically insignificant ($p = 0.155$). This supports the causal interpretation that household formation pressure primarily impacts rent growth with a lag, consistent with slow-moving supply constraints and lease adjustments.

TABLE 4 Placebo Test: RDI and Same-Year Rent Growth

	Coef.	Std. Err.	p-value	95% CI
RDI	-0.0122	0.0086	0.1554	[-0.0291, 0.0046]
Population Growth	0.0118	0.0020	0.0000	[0.0080, 0.0157]
Housing Starts (%)	-0.1777	0.0407	0.0000	[-0.2575, -0.0979]
Constant	0.0287	0.0201	0.1526	[-0.0106, 0.0680]
Observations	792			
F-statistic (first stage)	48.18			

We also implement a robustness check using a *lagged* version of the instrument: percent of 20-year-olds in year $t - 1$. Results, presented in Table 5, are nearly identical in sign and magnitude to the main specification, with the RDI coefficient remaining negative and significant at the 1% level. This helps rule out concerns of simultaneity and supports the argument that the instrument captures anticipated household formation shocks.

TABLE 5 Robustness: Lagged IV for RDI

	Coef.	Std. Err.	p-value	95% CI
RDI	-0.0288	0.0103	0.0050	[-0.0489, -0.0087]
Population Growth	0.0098	0.0019	0.0000	[0.0061, 0.0135]
Housing Starts (%)	-0.2529	0.0669	0.0002	[-0.3840, -0.1219]
Constant	0.0705	0.0239	0.0031	[0.0238, 0.1173]
Observations	693			
F-statistic (first stage)	30.27			

4.5 | Interpretation

Together, these results provide strong evidence that exogenous variation in household crowding pressureproxied by the size of the 20-year-old cohortcausally increases future rent growth. The absence of an effect on contemporaneous rents, and the consistency of estimates across lagged specifications, reinforce the temporal logic of the identification strategy and support the claim of a one-directional, crowding-to-rent causal channel.

While the IV model provides a causal estimate of the effect of crowding pressure on rent growth, the subsequent tests in Section 5 evaluate whether RDI and its change ΔRDI can serve as a useful predictive signals. These forecasting models do not claim to identify structural relationships, but rather assess the practical value of RDI for anticipating future market conditions.

5 | FORECASTING VALIDATION AND REAL-WORLD PERFORMANCE

This section evaluates whether RDI functions as a useful predictive signal for rent growth, independent of causal identification. We organize the analysis into five complementary tests: (1) a simple group-wise ANOVA showing forward rent segmentation by RDI level and movement; (2) an event study evaluating rent dynamics before and after regime shifts in RDI; (3) a top-vs-bottom quintile spread comparison across forecasting models; (4) a long-horizon signal using count-based RDI persistence;

and (5) a diagnostic of supply growth in context, showing how RDI mediates the impact of new supply. Together, these tests examine the predictive performance, durability, and explanatory clarity of RDI across diverse empirical contexts.

5.1 | Predictive Segment Testing: ANOVA of RDI

We test whether RDI and its delta can segment markets into meaningful groups in terms of future rent performance. Specifically, we split the samples into four regimes based on the two variables (greater/less-than-median RDI and positive/negative ΔRDI). We then examine whether average next-year relative real rent growth differs significantly across these groups.

Table 6 shows that dense markets have higher rent growth. Specifically, markets with RDI above the national median (2.4) experience average next-year real rent growth of 30 basis points, while markets with RDI below the median exhibit an average decline of 23 basis points. The difference is statistically significant (p$\leq 3.07^{-7}$), as confirmed by a one-way ANOVA.

Table 7 shows that as markets densify, their rent growth declines. Specifically, markets and years that have an increase in RDI (representing a growth in number of people per renter household) also have declines in rent of, on average, 31 basis points. This is significantly different (p$\leq 7.75^{-11}$) from markets that are de-densifying, which have rent growth, on average, of positive 37 basis points.

Grouped together, table 8 the difference between dense-but-dedensifying markets and sparse-but-densifying markets is 144 basis points.

TABLE 6 ANOVA: Real Rent Growth Next Year by RDI Level

Source	Sum of Squares	df	F	p-value	Mean Rent Growth
RDI Level	2,410,140	1.0	26.31	3.07e-07	RDI≥median: 30 bps RDI<median: -23 bps
Residual	311,308,300	3398			

TABLE 7 ANOVA: Real Rent Growth Next Year by ΔRDI

Source	Sum of Squares	df	F	p-value	Mean Rent Growth
RDI Growth Direction	3,883,449	1.0	42.59	7.75e-11	ΔRDI < 0: 37 bps ΔRDI ≥ 0: -31 bps
Residual	309,835,000	3398			

TABLE 8 ANVA and Group Means: Real Rent Growth Next Year by Combined RDI Level and Growth

Source	Sum of Squares	df	F	p-value	Mean Rent Growth
Combined Group	7,536,939	3.0	27.87	8.52e-18	RDI ≥ median ∩ ΔRDI < 0 : 73bps RDI ≥ median ∩ ΔRDI ≥ 0 : -2bps RDI < median ∩ ΔRDI < 0 : 10bps RDI < median ∩ ΔRDI ≥ 0 : -71bps
Residual	306,181,500	3396			

These results show that even without instrumentation, RDI and RDI growth serve as a powerful signals for forward rent performance. Practitioners can use RDI segmentation as a lightweight, interpretable classification rule for identifying tight rental markets.

5.2 | Event Study of RDI Regime Transitions

We next evaluate how rent growth responds dynamically to transitions into and out of RDI-growth regimes. Specifically, we examine rent growth before and after a market switches into a state of positive ΔRDI (increasing crowding) or negative ΔRDI (declining crowding). Here we examine results in terms of real rent growth relative to the mean rent growth of that year, to avoid capturing rent changes due to macro conditions.

Figure 4 illustrates rent growth over the two years ante and post regime-change. The green bars represent markets and periods where RDI switched from at least 3 years of consolidation to expansion (densification shifting to de-densification). The blue bars represent occasions when RDI switched from at least 3 years of densification to de-densification (from crowding to spreading).

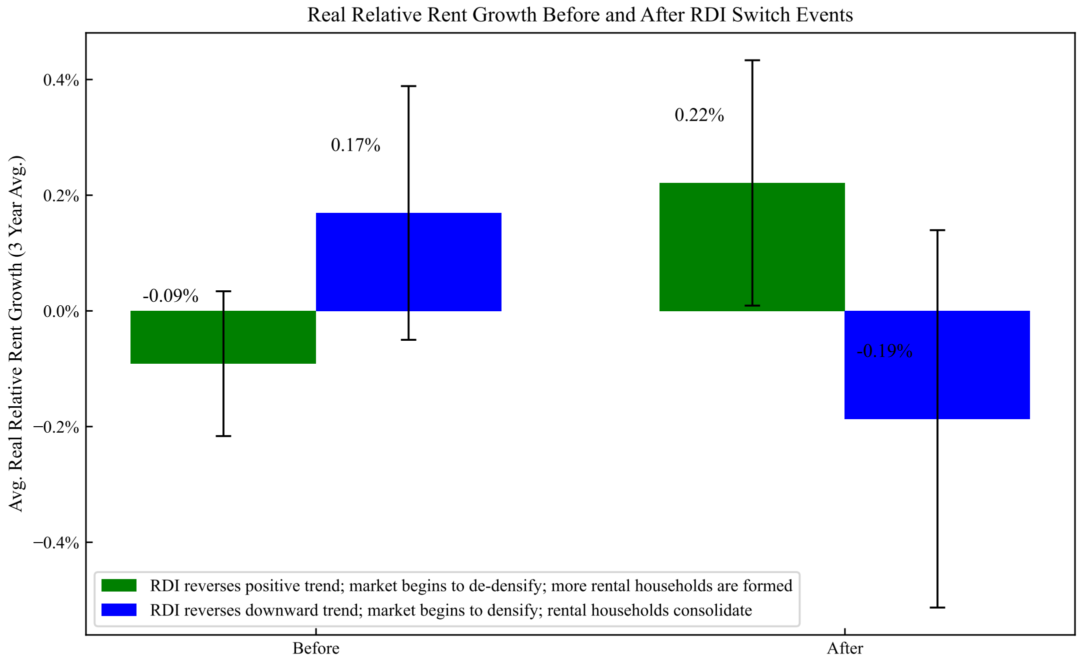
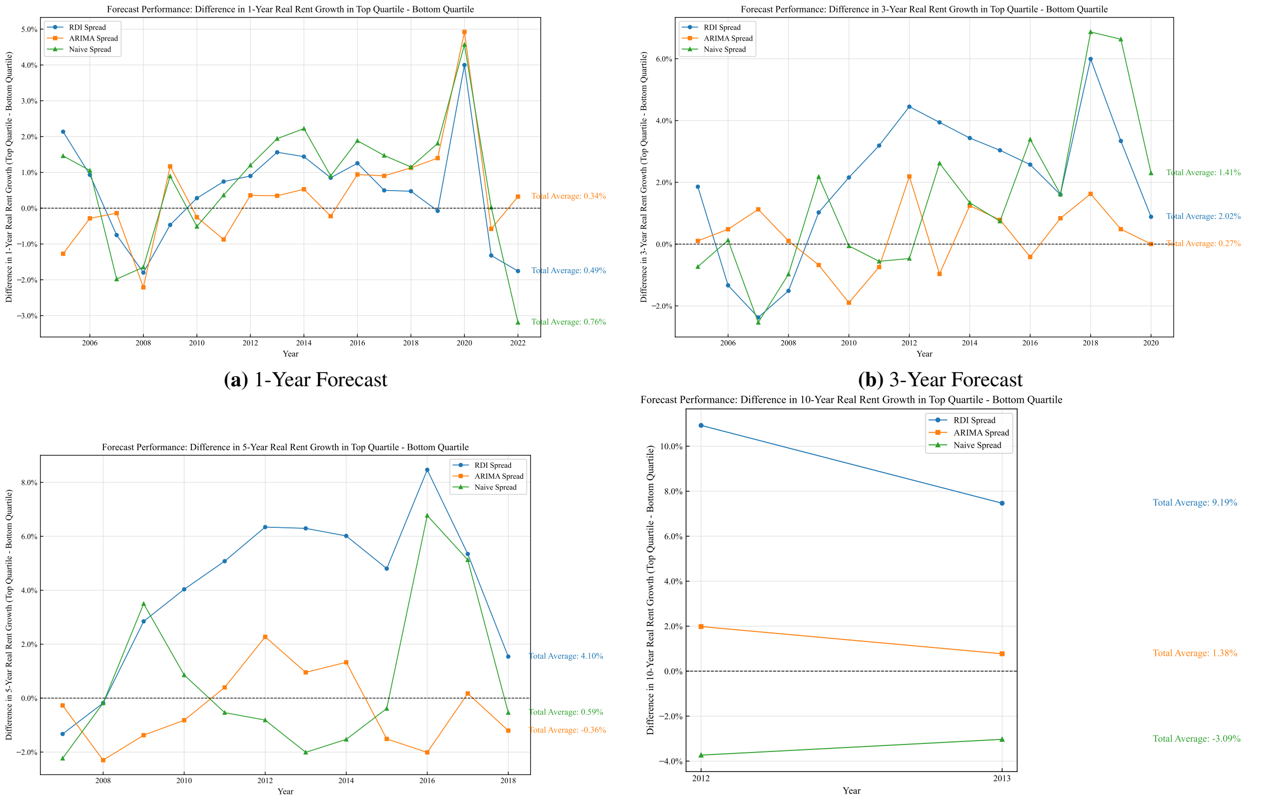


FIGURE 4 Real Relative Rent Growth Before and After a Market Switches From/To Positive/Negative ΔRDI

When markets reverse their positive RDI trend (crowding) and begin to form households, real relative rent growth shifts from negative to positive. The markets, on average, shift from a three year stretch of underperformance to a three year stretch of outperformance. The difference of 31 basis points per year is statistically signficant ($p < 0.013$). Conversely, when markets reverse their negative RDI trend (household formation) and begin to consolidate, real relative rent growth shifts from positive to negative. This shift however is not statistically significant ($p < 0.066$). These dynamics confirm that the onset of renter de-densification, as measured by RDI change, precedes statistically and economically meaningful changes in rent performance. RDI transitions can therefore serve as timely indicators of shifts in market pricing power.

5.3 | Forecast Spread Comparison: RDI vs ARIMA vs Naïve

We compare the predictive strength of the RDI against ARIMA-based and naive trailing average models in forecasting real rent growth across multiple time horizons. For each method, for each year, we group MSAs into quartiles based on their predicted growth (predicted by the RDI value, ARIMA, or naive method). We then compute the realized top-minus-bottom quartile spread. Here we limit the data set to the top 100 markets, as the ARIMA estimators did not converge on a large number of smaller markets. Figure 5 displays forecast accuracy over one-year, three-year, five-year, and ten-year windows. The RDI-based approach delivers stronger spread segmentation, particularly at longer horizons where traditional time-series methods tend to degrade. Forecasting one year forward based on the RDI of the prior year, the RDI method underperforms the Naive method slightly (30 basis points). In three-year forecasts, the RDI spread outperforms the next closest predictor by 60 basis points. At 5 years, the RDI outperforms by over 500 basis points. At ten years it outperforms by nearly 800 basis points.



(a) 1-Year Forecast

(b) 3-Year Forecast

(c) 5-Year Forecast

(d) 10-Year Forecast

FIGURE 5 Top-minus-Bottom Quintile Rent Growth Spread by Forecast Method and Horizon

These results reinforce the value of RDI as a forward-looking, long lived predictive signal. Forecasting long-term rent growth is notoriously difficult; most models lose power beyond a few years. That the RDI-based approach maintains signal strength across one-, three-, five-, and ten-year periods underscores its robustness. While time series models rely on trailing trends, RDI captures forward-looking, structural occupancy pressuremaking it an effective long-horizon indicator of underlying demand.

5.4 | RDI Positivity and National Rent Growth

To this point, we have examined the effects of RDI and RDI growth on an MSA level. Here, we consider the impact of RDI change on national rent growth. For each year we calculate the percent of markets (out of 200) that have positive RDI growth. The values range from a low of 21% in 2006, to 67.5% in 2022. For each of those years we calculate the mean real rent growth of the following year.

Figure 6 shows a consistent, robust, inverse relationship between number of RDI-increase markets and rent growth. Even with the effects of COVID (2020, 2021) the RDI-positive percent explains 32% of the variance in annual, national, real rent growth. Noteworthy is the 50th percentile on the x-axis, above which rent tends to be negative (5 of 8 years), and above which rent tends to be positive (7 of 9 years). Said differently, when most markets have renters who are crowding (experiencing positive RDI change) the national rent growth is likely to fall in the following year. Conversely, when most markets have renters who are forming households (experiencing negative RDI change), then the demand for rentals is likely to put upward pressure on national real rents.

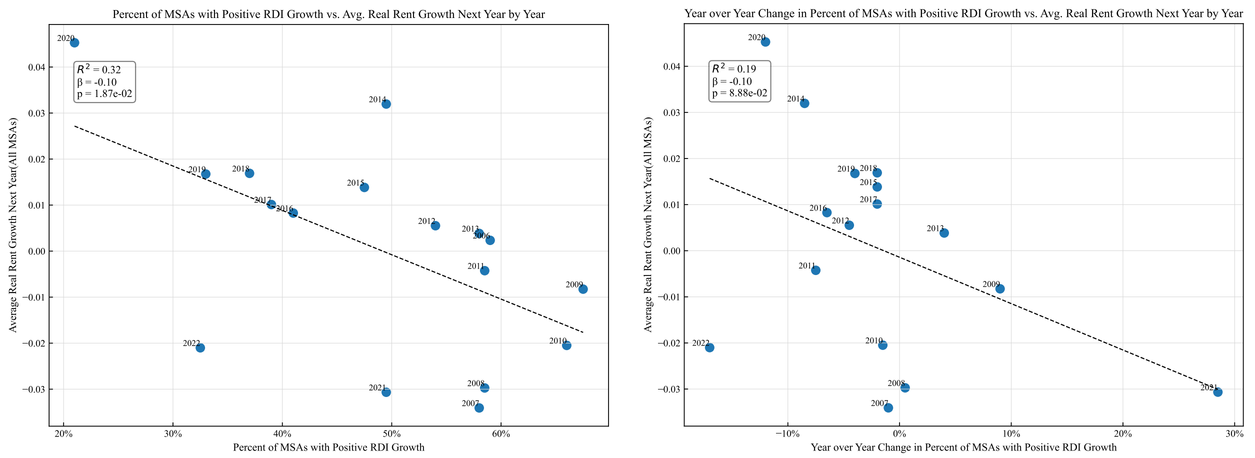


FIGURE 6 Impact of Positive RDI Percentage, and Change in Positive RDI Percentage, on the following year's real rent growth

Although the residuals are not autocorrelated and the regressors are stationary, to reduce the risk of spurious correlation, we take the first difference of the x variable and regress it against the real rent growth in the next year. The result is an an equally signed relationship with statistical significance. These results extend naturally from the evidence shown at the MSA level, and they reinforce the practical utility of RDI: even without complex forecasting models, the simple count of positive RDI signals—interpreted as consistent crowding pressure—can help periods of national demand pressure.

5.5 | Supply Growth in Context: RDI as a Mediating Lens

It is often assumed that elevated supply growth should depress rents in subsequent years. To test this assumption, we identify each MSA's year of maximum supply growth and examine the subsequent years real rent growth. Here we opt for real rent growth because the years of maximum supply are not the same across MSAs. Figure 7a shows that the relationship between supply growth and rent growth in the next yearis weak: 52% of markets experienced *positive* rent growth after their peak-supply year.

In contrast, when we introduce RDI in those same years, we gain valuable insight. We compute a simple measure of demand relative to supply of the form

$$\text{RDI\_of\_Supply}_{i,t} = \Delta\text{RDI}_{i,t} + \frac{\Delta\text{Supply}_{i,t}}{\text{RDI}_{i,t}}$$

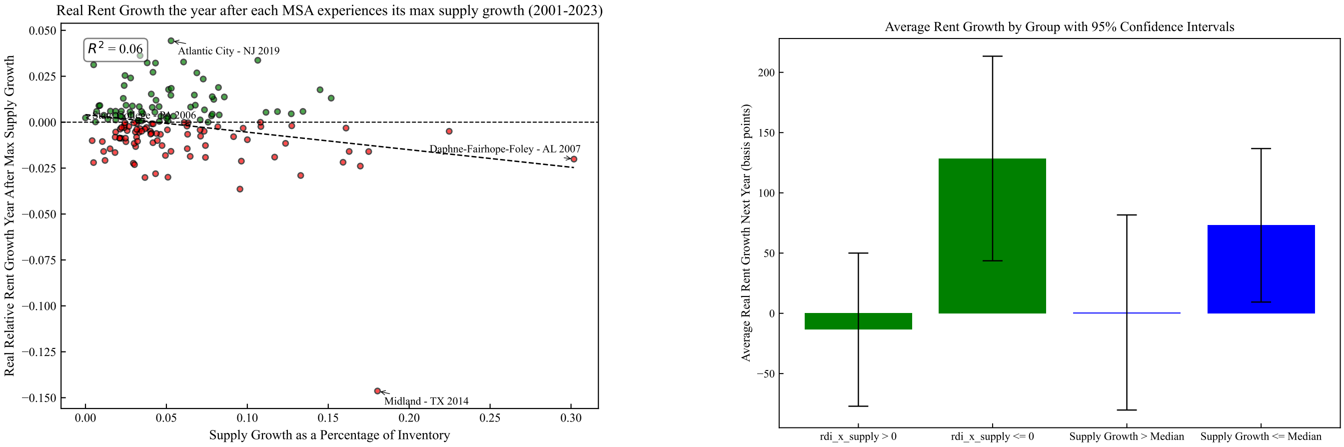
Where: i is an MSA in year t

Intuitively, this converts supply growth to a population-applicable figure, to which we add RDI-growth (negative in times of expansion; positive in times of crowding). As (Figure 7b) shows, when this value is greater than zero, rent in the following year is, on average, negative. When this value is less than zero, the next year's rent, on average, is positive. The difference between the two segments is 142 basis points, and the means are significantly different ($p < 0.0095$). This suggests that crowding pressures, not raw supply levels, capture more of the variance in subsequent rent performance. Even in high-supply environments, strong demand (captured by RDI) can offset what might otherwise appear to be oversupply conditions.

These results challenge the assumption that supply growth alone determines rent dynamics. Without considering how supply is absorbed relative to household formation, analysts risk misclassifying healthy markets as oversupplied. RDI offers a more reliable, demand-sensitive lens.

5.6 | Summary: RDI as a Practically Useful, Predictively Durable Demand Metric

The empirical tests in this section demonstrate that RDI—a simple, observable measure of rental household crowding—delivers meaningful insight into rental market behavior. Across multiple frameworks, RDI consistently outperforms:



(a) Supply Growth vs. Subsequent Rent Growth

(b) Supply Growth vs. RDI Growth

FIGURE 7 Crowding Absorption During Maximum Supply Years

- RDI group (above/below median) combined with RDI growth (positive/negative)classify markets into rent-growth and rent-decline groups, with a symmetric and significant difference of 144 basis points.
- RDI reversal signals periods of coming rent-growth change
- RDI value forecasts 3, 5, and 10-year future rent growth significantly better than ARIMA and Naive models.
- RDI values at the national level forecast next year explain a high percentage of the next year's real rent growth
- When RDI is compared with supply growth figures it effectively classifies over supplied markets

These results complement the causal identification strategy in Section **??**, where instrumented RDI growth was shown to drive future rent increases. Here, we show that RDI also performs well in practice without instrumentation, making it suitable for real-time forecasting, benchmarking, and investment decisions.

While RDI is not without limits—its precision may weaken in data-sparse MSAs or highly regulated rent environments—it stands out as a behaviorally grounded, interpretable, and statistically durable signal of latent rental housing demand.

6 | DISCUSSION

This paper demonstrates that crowding pressure, as measured by the Rental Density Index, serve as a robust and forward-looking indicator of rental market tightness. Our empirical analyses show that RDI reliably predicts future rent growth, highlighting the hidden intensity of rental housing demand even when traditional market metrics such as occupancy rates remain stable. By distinguishing between the causal role of RDI in driving rent growth and its practical forecasting utility, we establish the broad applicability and effectiveness of this indicator. The findings underscore its versatility as both a rigorous economic indicator of underlying market dynamics and a practical forecasting tool for diverse market participants.

6.1 | Advantages

Unlike traditional, static metrics such as occupancy and absorption, RDI dynamically captures behavioral adjustments by households responding to housing constraints, such as increased roommate formation or delayed household formation. These early behavioral signals often precede rent increases, positioning RDI as a valuable leading indicator.

Significantly, RDI retains predictive strength across various forecasting horizons, consistently outperforming traditional ARIMA and naive trailing-average models, even in out-of-sample tests. Its predictive accuracy, particularly notable in forecasting rent trends over longer periods (such as ten-year horizons), underscores its robustness and practical applicability.

The clear implications for policy-makers, investors, and renters make the variable particularly valuable. For policymakers, it offers an early warning signal of emerging housing shortages or surpluses, facilitating timely and targeted interventions such as zoning reforms, targeted subsidies, and expedited permitting processes. Investors and developers can integrate RDI into their risk management and feasibility analyses, enhancing strategic decisions about market entry, asset allocation, and development timing. Renters benefit from clearer market signals about impending rent increases, enabling better-informed housing decisions and advocacy efforts. Thus, RDI provides a structurally sound, scalable, and practical framework for anticipating and managing market dynamics across multiple contexts.

6.2 | Limitations: Spatial Spillovers

The current analysis treats each metropolitan statistical area (MSA) as an independent unit. In practice, however, housing market conditions are rarely contained by administrative boundaries. Migration, affordability spillovers, and commuting patterns often link neighboring MSAs, meaning that crowding pressures in one region may affect rental markets in adjacent areas. Ignoring these spillover dynamics may understate the correlated nature of demand shifts, especially in highly integrated urban corridors. Incorporating spatial dependence structures or spatially lagged RDI variables in future work may enhance model precision and improve detection of regional displacement or diffusion effects.

While we do not formally estimate a spatial lag model, we assess geographic clustering by examining pairwise correlations in ΔRDI across neighboring MSAs. We find strong co-movement among regionally linked marketse.g., New York and Northern New Jersey ($r = 0.70$), San Francisco and San Jose ($r = 0.69$), and DallasFort Worth and Austin ($r = 0.77$). More details can be found in Appendix A. These results suggest that crowding pressures often evolve in tandem across proximate metros, reflecting regional migration flows, shared housing constraints, or price spillovers. Future work could formalize these relationships using spatial dependence structures or migration-network adjacency matrices.

7 | CONCLUSION

We propose the Rental Density Index (RDI) as a demand-centric metric for measuring latent pressure in multifamily rental markets. By tracking changes in population per occupied unit, RDI captures household-level crowding dynamics that precede and predict future rent movement. Our empirical results, validated across causal models, forecasting tests, and event studies, establish RDI as both a leading indicator and a practical classification tool.

The simplicity of RDI is a strength. While traditional demand estimation often requires detailed demographic or income data, RDI can be computed with widely available population and housing stock data. Yet it performs comparablyif not betterthan more complex models, especially over long forecast horizons. It is also behaviorally grounded: renters adjust space usage when pricing becomes constrained, and this adjustment leaves measurable traces in the data.

Future research may extend this work in several directions. RDI could be evaluated alongside migration flows, income segmentation, or micro-level leasing data to refine its predictive specificity. It may also prove useful in pricing models, rent control policy evaluation, or early-warning systems for affordability crises.

In sum, this paper offers both a new way to quantify housing demand and a practical framework for translating that signal into market insight. RDI helps bridge the gap between structural economics and applied decision-makingand in doing so, offers analysts, policymakers, and investors a grounded, interpretable, and scalable tool for anticipating market behavior.

APPENDIX A: SPILLOVER EFFECT CORRELATION MATRIX

TABLE 9 Cross-MSA Correlation in ΔRDI Among Regional Pairs

MSA 1	MSA 2	Correlation (r)
New York, NY	Northern New Jersey, NJ	0.70
San Francisco, CA	San Jose, CA	0.69
DallasFort Worth, TX	Austin, TX	0.77
Los Angeles, CA	Inland Empire, CA	0.52
Miami, FL	Palm Beach, FL	0.54

AUTHOR CONTRIBUTIONS

All authors contributed equally

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FINANCIAL DISCLOSURE

CONFLICT OF INTEREST

DATA DISCLOSURE

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